3(c)

MLE objective ~~of linear regression~~ in heteroscedastic setting

$$= \underset{w}{\arg\min}\ \frac{1}{2}\left\{\frac{\alpha_n (t_n - w(\phi(x_n)))^2}{\sigma_n^2}\right\}$$

$$= \underset{w}{\arg\min}\ \frac{1}{2}\left\{r_n (t_n - w^T(\phi(x_n)))^2\right\}$$

where weighting factor $r_n = \frac{\alpha_n}{\sigma_n^2}$ for n^{th} data point

Hence the MLE estimate reduces to minimization of equation 3 hence sum of squares error function is..

$$E_D(w) = \frac{1}{2}\sum_{n=1}^{N} r_n (t_n - w^T(\phi(x_n)))^2$$

(ii) to minimize $E_D(w)$ we find $\nabla_w E_D(w)$

$$\nabla_w E_D(w) = 2 \cdot \frac{1}{2}\sum_{n=1}^{N} r_n (t_n - w^T(\phi(x_n)))(-\phi(x_n))$$

$$= -\sum_{n=1}^{N} r_n t_n \phi(x_n) + \sum_{n=1}^{N} r_n w^T \phi(x_n) \cdot \phi(x_n)$$

putting $\nabla_w E_D(w) = 0$ we get

$$\sum_{n=1}^{N} r_n W^T \phi(x_n) . \phi(x_n) = \sum_{n=1}^{N} r_n t_n \phi(x_n)$$

$$\Rightarrow \sum_{n=1}^{N} r_n \phi(x_n)^T w \phi(x_n) = \sum_{n=1}^{N} r_n t_n \phi(x_n)$$

$$\Rightarrow W = \frac{\sum_{n=1}^{N} r_n t_n \phi(x_n)}{\sum_{n=1}^{N} r_n \phi(x_n)^T \phi(x_n)}$$

this value of w minimizes the equation